

(Recall) last time we described the following correspondences

Frieze patterns of order n $\longleftrightarrow Gr_{2n}$

$\nwarrow$ bijection

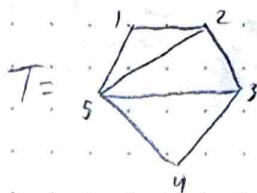
$$\left\{ (\Delta_{ij})_{1 \leq i < j \leq n} \begin{array}{l} 1) \Delta_{ij} \text{ satisfy 3-term Plücker rels.} \\ 2) \Delta_{ij} > 0. \end{array} \right\}$$

* Integer frieze patterns correspond to triangulations of n -gons

Fact: the number of triangulations of an n -gon = C_{n-2} $\swarrow$ Catalan #

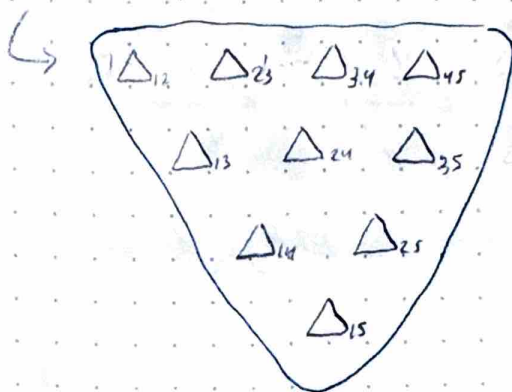
EX: let $n=5$

$\rightarrow$ use 3-term Plücker relation

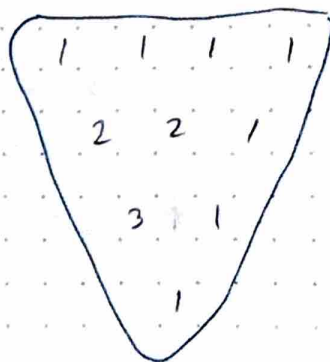


$$\Delta_{12} = \Delta_{23} = \Delta_{34} = \Delta_{45} = \Delta_{15} = 1$$

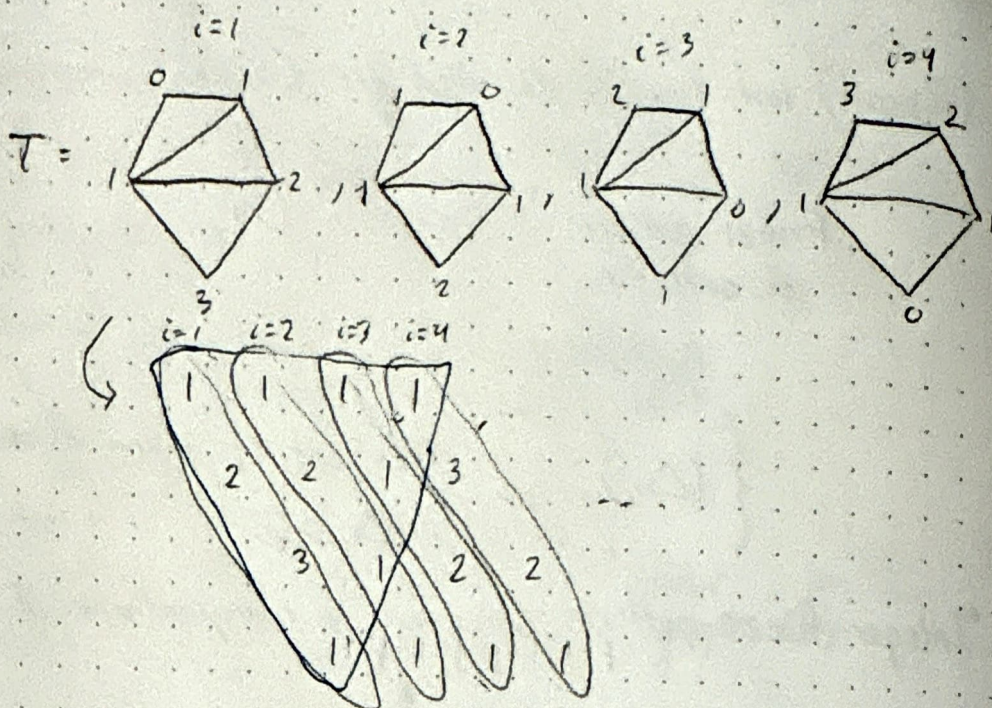
$$\Delta_{25} = \Delta_{35} = 1, \Delta_{13} = \Delta_{24} = 2, \Delta_{14} = 3$$



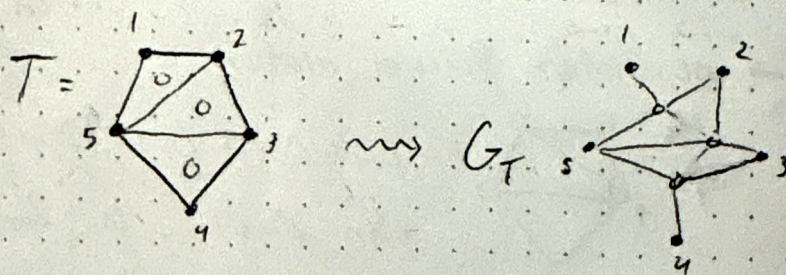
$=$



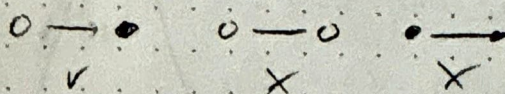
→ using the Conway-Coxeter construction



→ using Plabic Graphs (non-recursive)



(edges drawn between opposite color vertices)



then, $\Delta_{ij} = \# \text{ perfect matchings of } G_T \setminus \{ \text{vertices } i \text{ and } j \}$

Exercise: show that the three constructions give the same result for any fixed triangulation T .

(Resource) Postnikov, Farber — "Arrangements of equal minors"

↳ how can we describe possible collections of pairwise equal Plücker coordinates Δ_I for $Gr_{k,n}$, assuming that all $\Delta_I > 0$.

Def'n

↳ an arrangement of smallest minors is a subset $S \subseteq \binom{[n]}{k}$, s.t.

→ $\exists (\Delta_I) \in Gr_{k,n}$ with $\Delta_I = 1$ for all $I \in S$.

(and) → for all $\Delta_J > 1$ for all $J \notin S$.

similarly, an arrangement of largest minors is a subset $L \subseteq \binom{[n]}{k}$, s.t.

→ $\exists (\Delta_I) \in Gr_{k,n}$ with $\Delta_I = 1$ for all $I \in L$.

(and) → $0 < \Delta_J < 1$ for all $J \notin L$.

Consider the case $k=2$, and identify a subset $S \subseteq \binom{[n]}{2}$ with a subgraph $G \subseteq K_n$ w/ edge set S .

→ a planar graph is a graph that can be drawn on the plane w/o crossing edges

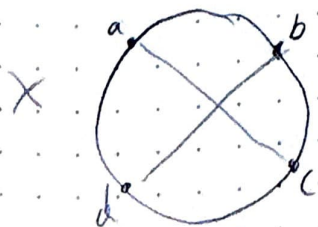
→ Conway's thrackles are graphs drawn on the plane s.t. any two edges have exactly one common point.

Conjecture (Conway):

↳ for all thrackles, $\# \text{ edges} \leq \# \text{ vertices}$ — equality if we have a maximal thrackle

Def'n (outer planar graph):

↳ graph $G \subseteq K_n$ with no crossing edges $(a,c) \nmid (b,d)$, assuming that all vertices are arranged on the circle in clockwise order.



for $a < b < c < d$

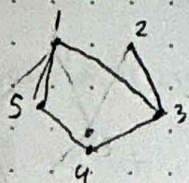
is a graph
 an outer thrackle $G \subset K_n$ such that any pair of edges cross or have a common vertex when all vertices are arranged on the circle.

Thm:

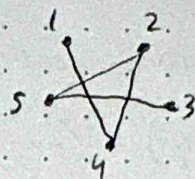
↳ for $k=2$, the arrangement of smallest minors are exactly outer planar graphs.

Arrangements of largest minors are exactly outer thrackles.

Ex:



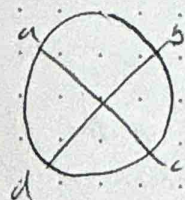
outer planar graph



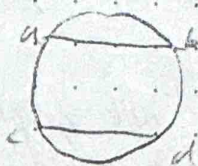
an outer thrackle

(one direction of the proof involves 3-term Plücker relations.

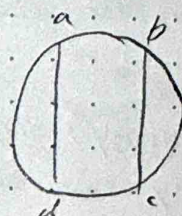
↳ let $a < b < c < d$, $\Delta_{ac} \cdot \Delta_{bd} = \Delta_{ab} \cdot \Delta_{cd} + \Delta_{ad} \cdot \Delta_{bc}$



=



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Claim: maximal outer planar graphs are exactly triangulations of n -gons, and thus there are C_{n-2} of them on n vertices.

Exercise: show that maximal outer thrackles have exactly n edges.

↳ calculate the number of such thrackles

↳ view as abstract subgraphs of K_n

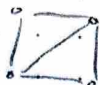
Consider the following examples

triangulations:

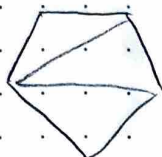
$n=3$



$n=4$



$n=5$

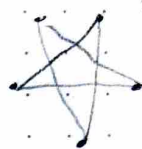
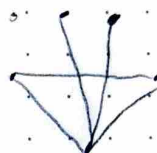
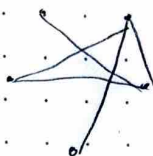


+ rotations

outer thrackles



+ rotations



+ rotations